CHERN CLASSES OF A PERFECT OBJECT ARE CENTRAL AND COMMUTE WITH BIVARIANT CLASSES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Vanishing criteria for bivariant classes, and the computation of Chern classes after a proper birational modification. Same chapter chow.tex, labels lemma-bivariant-zero (proof line 9409) and lemma-chern-classes-computed (proof line 9411, claimed as S14-algebra-chern-classes-computed). The first reduces the vanishing of the difference of two composites to the case of an integral base, and the second replaces the object by a bounded complex of finite locally free modules of constant rank on a proper birational model.
- Centricity and commutation for Chern classes of locally free modules, and the construction of Chern classes as polynomials in them. Same chapter chow.tex, label lemma-cap-commutative-chern (proof line 9421, claimed as S14-algebra-cap-commutative-chern). Gives the commutation for each of the locally free terms, whence for the polynomial expression defining the Chern class of the complex.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Navigation pointer for \ref{lemma-bivariant-zero} used inside the counted span (line 9410); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UA: sources/stacks-chow/pinned-original-source.tex, source lines 7581–7581. Navigation pointer for \ref{lemma-cap-commutative-chern}

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used inside the counted span (line 9421); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GUF: sources/stacks-chow/pinned-original-source.tex, source lines 9356–9356. Navigation pointer for \ref{lemma-chern-classes-computed} used inside the counted span (line 9411); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 9395); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $E \in D(\mathcal{O}_X)$ be perfect. If the Chern classes of E are defined then*

1. *$c_p(E)$ is in the center of the algebra $A^*(X)$, and*
2. *if $g : X' \to X$ is locally of finite type and $c \in A^*(X' \to X)$, then $c \circ c_p(E) = c_p(Lg^*E) \circ c$.*

Proof. Part (1) follows immediately from part (2). Let $g : X' \to X$ and $c \in A^*(X' \to X)$ be as in (2). To show that $c \circ c_p(E) - c_p(Lg^*E) \circ c = 0$ we use the criterion of Lemma 02UC. Thus we may assume that X is integral and by Lemma 0GUF we may even assume that E is represented by a bounded complex $\mathcal{E}^\bullet$ of finite locally free $\mathcal{O}_X$-modules of constant rank. Then we have to show that

$$c \cap c_p(\mathcal{E}^\bullet) \cap [X] = c_p(\mathcal{E}^\bullet) \cap c \cap [X]$$

in $\mathrm{CH}_*(X')$. This is immediate from Lemma 02UA and the construction of $c_p(\mathcal{E}^\bullet)$ as a polynomial in the chern classes of the locally free modules $\mathcal{E}^n$. □